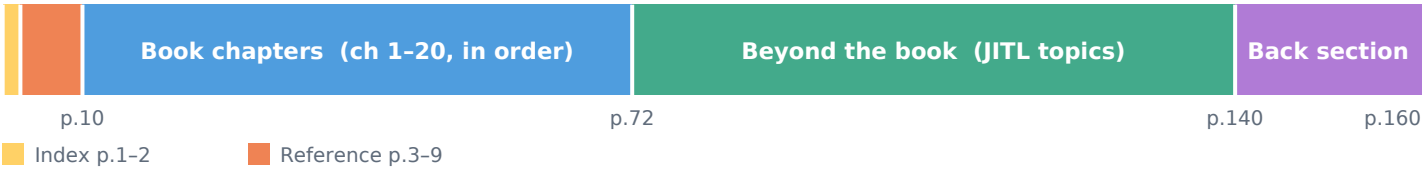
CCNA Notebook Guide

How to set up and fill the pen+GEAR 1-subject notebook (80 sheets = 160 pages, college ruled)
Method: Cornell notes · one topic per two-page spread · cues are questions · summary after the lab

1 Page map: decide where everything goes before you write



Pages	Section	What goes there
1–2	Index	Topic → page number. Fill it in as you go. (Mirror in the app: Book → Notebook.)
3–9	Reference	pp.3–8: subnetting, ports, AD/route codes, IOS modes, "book is outdated" list, OSI chart (pages 4–5 of this guide). p.9: your tab-colour key.
10–71	Book chapters	One spread (2–4 pages) per chapter you read or skim, starting on an even page. Skip chapters 4, 14 and Appendix A.
72–139	Beyond the book	VLANs, STP, EtherChannel, OSPF, IPv6, FHRP, NAT, services, QoS, security, wireless, automation
140–160	Back section	Start on p.160 and work backwards: mistake log, weekly review log (page 6)

2 Setup tonight (≈20 min)

1. Write the page number in the corner box of every page, 1–160. Odd numbers are right-hand pages, even numbers left-hand.
2. Pages 1–2: write INDEX at the top. Draw 3 columns: Topic | Page | Tab.
3. Pages 3–8: write the reference titles (see pages 4–5). Page 9: copy the tab-colour key from box 3.
4. Rule the first chapter spread (pages 10–11) as shown on page 2. Every spread starts on an even page.
5. Page 160: write MISTAKE LOG. Page 150: WEEKLY REVIEW.
6. Stick a tab on p.3 (Reference) and p.160 (Mistakes).
7. Don't use the perforations. Torn-out pages break the index.

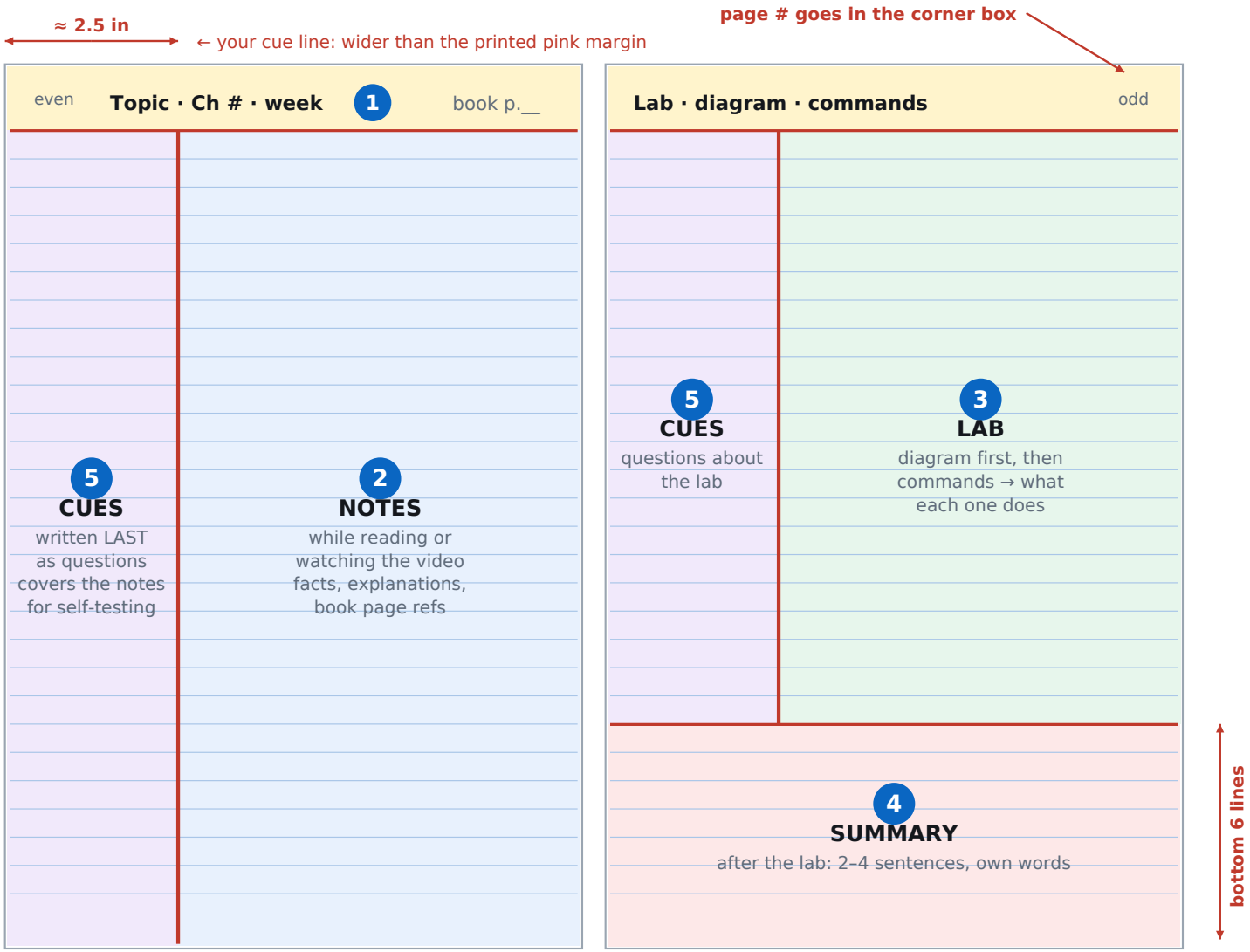
3 Sticky-tab colour key

One colour per exam domain. Tab the first spread of each domain. Colour each box with the tab you choose:

- ☐ Network Fundamentals — exam 20%
- ☐ Network Access — exam 20%
- ☐ IP Connectivity — exam 25%
- ☐ IP Services — exam 10%
- ☐ Security Fundamentals — exam 15%
- ☐ Automation & Programmability — exam 10%
- ☐ Reference pages
- ☐ Mistake log

How to rule each topic spread

Draw these lines in pen (red below) on every new topic. Each colour zone has one job, and you fill them in this order: 1 → 5.



The 5 zones

1 Header line
Topic, chapter, week and book page, written beside the corner box (the page number goes in the box). Lets the index point here.

2 Notes (left page)
While reading or watching. Short phrases, arrows, your own abbreviations. Leave gaps to add to later.

3 Lab (right page)
Draw the topology first. Then every command you ran, with what it does in plain words.

4 Summary
Write it only after the lab, not after the video. If you can't summarise it, redo the lab.

5 Cues (both pages)
Last step. Turn each block of notes into a question. Cover the notes and answer out loud.

— = lines you draw yourself (pen + ruler, or the edge of the book)

Big topic (e.g. OSPF)? Continue onto the next spread and keep the same layout. Only the last right page gets the summary box.

Worked example: Chapter 2, encapsulation

Blue = notes and lab. Purple = cues (written last). Red = the lines you drew. Your handwriting will be bigger: let it spill onto a second spread.

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Ch 2 · Encapsulation · wk 1 *book p.__*

5 steps of encapsulation?

SENDING side: 5 conversion steps

1. Build data (app: e-mail text)

2. Package for transport → SEGMENT

```
TCP/UDP header + port numbers
```

Which layer adds src/dest IP?

3. Add network header → PACKET

```
src + dest IP → routers read this
```

Who adds MACs?

4. Add data-link header → FRAME

```
src + dest MAC + FCS trailer
```

Why rebuilt at every hop?

```
!! new frame at EVERY hop
```

5. Convert to BITS on the cable

PDU names L4 / L3 / L2 / L1?

PDUs: Data → Segment → Packet → Frame → Bits

"Don't Some People Fear Birthdays"

What happens on the receiver?

RECEIVING side = de-encapsulation

each layer strips its header,

passes the rest UP

OSI ↔ TCP/IP?

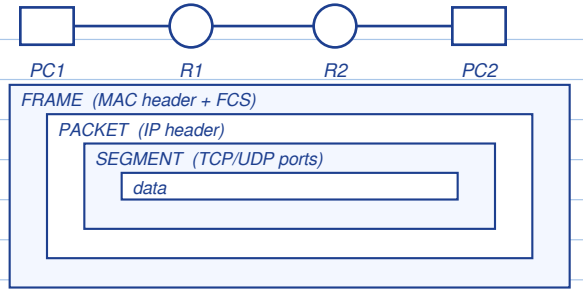
OSI 5-6-7 = TCP/IP Application

OSI 3 = Internet OSI 1-2 = Link

see also: Ch 9 (TCP), Ch 19 (ARP)

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Lab: watch encapsulation hop by hop



Which addresses change at a router?

app Lab 5: ping PC1 → PC2

```
ping 192.168.2.10
→ watch it hop PC1-R1-R2-PC2
R1# show arp
  = MACs R1 learned (L2 addresses)
```

IP src/dst = SAME end to end

MAC src/dst = rewritten by EACH router

SUMMARY

Each layer wraps the one above in its own header (encapsulation).

IP addresses get the packet across networks and never change.

MAC addresses only get it across ONE link, so every router strips the frame and builds a new one. Receiver unwraps in reverse.

Reference pages (notebook pp. 3–8)

Copy these in once, then add to them as you go. They are the pages you will flip to most.

p.3 Subnetting

```
Bits   128  64  32  16   8   4   2   1
Mask   128 192 224 240 248 252 254 255
```

/	Mask	Block (octet)	Hosts
/16	255.255.0.0	1 (2nd)	65,534
/17	255.255.128.0	128 (3rd)	32,766
/18	255.255.192.0	64 (3rd)	16,382
/19	255.255.224.0	32 (3rd)	8,190
/20	255.255.240.0	16 (3rd)	4,094
/21	255.255.248.0	8 (3rd)	2,046
/22	255.255.252.0	4 (3rd)	1,022
/23	255.255.254.0	2 (3rd)	510
/24	255.255.255.0	1 (3rd)	254
/25	255.255.255.128	128 (4th)	126
/26	255.255.255.192	64 (4th)	62
/27	255.255.255.224	32 (4th)	30
/28	255.255.255.240	16 (4th)	14
/29	255.255.255.248	8 (4th)	6
/30	255.255.255.252	4 (4th)	2

Block = 256 − mask octet. Network = multiple of the block ≤ host octet. Broadcast = next multiple − 1. Wildcard = 255 − mask octet.

p.4 Ports

TCP	Port	UDP	Port
FTP data / control	20 / 21	DNS (queries)	53
SSH	22	DHCP server / client	67 / 68
Telnet	23	TFTP	69
SMTP	25	NTP	123
DNS (zone transfer)	53	SNMP / traps	161 / 162
HTTP	80	Syslog	514
POP3 / IMAP	110 / 143	RIP	520
BGP	179	CAPWAP ctrl / data	5246 / 5247
HTTPS	443	RADIUS	1812 / 1813
TACACS+	49		

p.5 AD values & route codes

Source	AD	Code	Meaning
Connected	0	C	connected subnet
Static	1	L	local /32 (own IP)
eBGP	20	S / S*	static / default
EIGRP	90	O	OSPF
OSPF	110	D	EIGRP
RIP	120	R	RIP
iBGP	200	B	BGP
Unusable	255	[110/2]	[AD / metric]

Route choice: 1) longest prefix 2) lowest AD 3) lowest metric (ties = load-balance).

p.6 IOS modes & first commands

Prompt	Mode	Get in	Get out	Typical commands
R1>	User EXEC	(login)	exit	ping, traceroute, show (limited)
R1#	Privileged EXEC	enable	disable	show …, copy run start, debug, reload
R1(config)#	Global config	configure terminal	exit / end / Ctrl-Z	hostname, enable secret, ip route, vlan, router ospf
R1(config-if)#	Interface	interface g0/0	exit / end	ip address, no shutdown, description, switchport
R1(config-line)#	Line	line vty 0 4	exit / end	password, login (local), transport input ssh
R1(config-router)#	Routing protocol	router ospf 1	exit / end	network … area 0, router-id, passive-interface

Must-know show commands: show ip interface brief · show running-config · show ip route · show vlan brief · show interfaces trunk · show mac address-table · show cdp neighbors · show ip ospf neighbor · show access-lists

Reference pages (continued)

p.7 "The book is out of date" list

The Companion Guide is from 2003. Each time you hit a ⚠ in the app's chapter page, add a line here so you don't learn the old version.

Ch	The book says…	Today / the exam says…
3	WEP, LEAP for wireless security	Broken. WPA2 (AES) and WPA3 (SAE)
3	Pick straight-through vs crossover carefully	Auto-MDIX fixes it; still know the rule
6	10BASE5 / 10BASE2 coax, 802.1D STP	Gone. Rapid STP (802.1w / Rapid PVST+)
7	RARP and BOOTP; classful A/B/C networks	DHCP; CIDR (any prefix length)
8	Subnets = $2^n - 2$ (no subnet zero / all-ones)	Subnets = 2^n (ip subnet-zero default). Hosts still $2^h - 2$
10	ISDN, Frame Relay, dial-up WANs	Ethernet WAN, MPLS, broadband VPN, SD-WAN
11	HyperTerminal, setup mode	screen/minicom/PuTTY; answer "no" to setup
12	enable password, Telnet on VTY, serial clock rate	enable secret, SSH, no serial on v1.1
15–16	IGRP, RIP in depth, classful routing	IGRP gone; OSPF is what v1.1 tests
17	ICMP source quench	Deprecated

p.8 OSI ↔ TCP/IP chart

#	OSI layer	TCP/IP	PDU	Devices	Protocols / addresses
7	Application	Application	Data		HTTP, DNS, FTP, SMTP, SNMP, SSH
6	Presentation		Data		encryption, compression, formats
5	Session		Data		session setup / teardown
4	Transport	Transport	Segment	(firewall)	TCP, UDP, port numbers
3	Network	Internet	Packet	Router, L3 switch	IP, ICMP, OSPF, IP addresses
2	Data Link	Link	Frame	Switch, bridge, AP	Ethernet, 802.1Q, MAC addresses
1	Physical	Link	Bits	Hub, repeater, cable	UTP, fiber, radio

Mnemonics: top-down "All People Seem To Need Data Processing"; PDUs "Don't Some People Fear Birthdays" (Data, Segment, Packet, Frame, Bits).

The index (pp. 1–2) and the back section (pp. 140–160)

Index: what pages 1–2 look like

Topic	Page	Tab	Topic	Page	Tab
Ch 1 Binary, hex, IP & masks	10	Fund.	VLANs & trunks	72	Access
Ch 2 OSI, TCP/IP, encapsulation	12	Fund.	Spanning Tree / RSTP	76	Access
Ch 3 Media (skim)	16	Fund.	OSPF	84	IP Conn.
Ch 5 Ethernet frames	18	Fund.	IPv6	90	IP Conn.
…			…		

Use the left-hand (even) page number of each spread. Write the entry as soon as you start a spread, not at the end. Enter the same page number in the app (Book → chapter → Notebook page, or Book → Notebook for everything); it suggests the next free page.

Mistake log: start on p.160 and work backwards

One line per missed question from Boson, the app's chapter quizzes or Anki. The app keeps its own Weak topics list automatically; this log is for the misses you want to think through. Three misses on the same topic → redo that spread from memory.

Date	Source	Topic	What I got wrong	The right idea	Spread
11/14	Quiz ch 8	Subnetting	Said 6 subnets for 3 borrowed bits	$2^3 = 8$ (subnet zero is allowed)	p.24
11/16	Anki	Ports	TFTP = TCP 69	TFTP = UDP 69 (no handshake)	p.4

Weekly review log: start on p.150 and work backwards

Every Sunday at 4 pm, one row per week. Compare the hours with the plan's split (60/40 theory/practice in weeks 1–2, 40/60 after that, 20/80 in week 12).

Wk	Hours (theory / practice)	Chapters / topics done	Lab redone from memory	Weakest topic → next week
1	5 / 4	Ch 1, 2, 5	Lab 1 Hello, switch	hex conversions
2				
3				

Using the notebook every session

One study session, in order



Daily, 7:30 pm (10–15 min)

- Anki deck, then 10 min of subnetting (app → Drills).
- Glance at today's spread and answer the cues out loud.
- Log any misses in the back of the notebook.

Sunday, 4 pm weekly review

- Cover the notes column on every spread from this week and answer the cues.
- Redo one lab from memory, without the brief. Check it with the app's Check button.
- Fill one weekly-review row (p.150 backwards).
- Any cue you missed: rewrite that bit of the notes, or redo the spread.

Rules of thumb

- Facts go in Anki (port numbers, AD values). How things work goes in the notebook.
- Cues are questions, not facts: "Why does the frame change at each hop?", not "frames change".
- Diagrams beat paragraphs. Draw the topology for every lab.
- Commands always get a meaning: "no shutdown → bring the interface up".
- Leave a blank page at the end of each domain for later additions.
- Book chapters you skip (4, 14, App A) get no spread; their one-page summaries are in the app.
- Annotate the book lightly: mark skip/skim, flag ⚠ outdated bits, write the notebook page in the margin.

How the notebook and the app fit together

- Book tab → chapter: tick the reading guide, set the Book page and the Notebook page, take the chapter quiz.
- Book → Notebook: the digital copy of pages 1–2, with page numbers for every chapter and beyond-the-book topic.
- Cornell notes in the app = your cue questions, so you can self-test on the phone. Paper keeps the diagrams.
- Quiz misses land in Today → Weak topics automatically; copy the ones you need to think about into the mistake log.